

BRIDGING THE GAP: ADDITIVE STRUCTURE IN THE TWIN PRIME COFACTORS

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ABSTRACT. We introduce the Bridging Conjecture: for every K , there exist $a, b, k \in A002822$ with $a < K$, $b < K$, and $k = a + b > K$, where $A002822$ denotes the set of positive integers m such that $6m - 1$ and $6m + 1$ are both prime (twin prime cofactors). We show that the Bridging Conjecture is equivalent to the Twin Prime Conjecture assuming Dubner's Conjecture 1, and that it follows unconditionally from Dubner's Conjecture 3. As a corollary, the Bridging Conjecture implies $\pi_2(x) \gg \log x$, a conditional lower bound on the twin prime counting function derived purely from the additive structure of $A002822$. We also re-express Dubner's exception set A243956 in set-theoretic terms as $D = \mathbb{N}_{\geq 1} \setminus \{i + j : i, j \in A002822\}$, which explains the additive origin of its elements rather than merely listing them, and reformulates Dubner's Conjecture 3 as the assertion that D is finite.

1. INTRODUCTION

A *twin prime pair* is a pair of primes $(p, p + 2)$. The integer $p + 1$ between them is divisible by 6; write $p + 1 = 6m$, so m is a positive integer with $6m - 1$ and $6m + 1$ both prime. The set of such cofactors,

$$A002822 = \{m \in \mathbb{N} : 6m - 1 \text{ and } 6m + 1 \text{ are both prime}\},$$

is OEIS sequence A002822 [8]. The Twin Prime Conjecture asserts that $A002822$ is infinite. A prime belonging to a twin prime pair is called a *t-prime*.

In 1996, Stephen Wagler communicated a question to Paulo Ribenboim: what is the smallest middle number that is not the sum of two t-primes? Ribenboim asked Harvey Dubner to investigate [2, pp. 291–299]. Dubner's paper [1] is the primary published source; he formulated four conjectures in response, which we recall in Section 2. Wagler's question is closest in spirit to Conjecture 2, which asks whether every middle number greater than 6 is the sum of two t-primes. In Section 3 we re-express Dubner's exception set A243956 in set-theoretic terms as the complement of the sumset of $A002822$ in $\mathbb{N}_{\geq 1}$, which explains the

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additive origin of the exceptions rather than merely listing them, and reformulates Dubner's Conjecture 3 as the assertion that this complement is finite. In Section 4 we introduce the *Bridging Conjecture* and establish its relationships to the Twin Prime Conjecture and Dubner's conjectures.

2. DUBNER'S CONJECTURES

Dubner [1] stated four conjectures, which we number as he does.

Conjecture 2.1 (Dubner, Conjecture 1). Every middle number greater than 6 is the sum of two middle numbers.

In terms of $A002822$: a middle number $6m$ is the sum of two middle numbers $6a$ and $6b$ if and only if $m = a + b$ with $a, b \in A002822$. So Conjecture 2.1 is equivalent to: every $m \in A002822$ with $m > 1$ is the sum of two smaller elements of $A002822$. Dubner verified this up to 2×10^{10} .

Conjecture 2.2 (Dubner, Conjecture 2). Every middle number greater than 6 is the sum of two t-primes.

Conjecture 2.3 (Dubner, Conjecture 3). Every multiple of 6 greater than 4206 is the sum of two middle numbers.

Conjecture 2.4 (Dubner, Conjecture 4a). Every even number greater than 4208 is the sum of two t-primes.

Conjecture 2.3 is strictly stronger than Conjecture 2.1: it requires all sufficiently large multiples of 6 to decompose, not only those that are middle numbers. The sumset reformulation of Conjecture 2.3 is developed in Section 3.

3. THE SUMSET OF $A002822$ AND DUBNER'S EXCEPTION SET

The natural object here is the sumset of $A002822$ with itself. For any $i, j \in A002822$ the sum $i + j$ is a positive integer, and this sum need not itself belong to $A002822$: it may lie in $A002822$ (a middle number decomposition of $i + j$) or in $A067611$ (the complement). In either case $i + j$ is an element of $\mathbb{N}_{\geq 1}$ produced by the pair (i, j) . Iterating over all pairs in $A002822$ and collecting their sums therefore generates a subset of $\mathbb{N}_{\geq 1}$ with no restriction on whether the sums are twin-prime cofactors. Let D denote the complement of this sumset in $\mathbb{N}_{\geq 1}$, i.e. the set of positive integers admitting no such representation; this is OEIS A243956 [9]. Dubner identified eleven multiples of 6 that are not the sum of two middle numbers,

$$\{96, 402, 516, 786, 906, 1116, 1146, 1266, 1356, 3246, 4206\},$$

and conjectured these are the only exceptions. A243956 is derived from Dubner's set by dividing each element by 6 to recover the cofactor and adjoining 1 (which lies in A002822 but cannot be expressed as a sum of two elements of A002822), giving

$$D = \{1, 16, 67, 86, 131, 151, 186, 191, 211, 226, 541, 701\}$$

up to 2×10^{10} , and Conjecture 2.3 states that D is exactly this finite set. Conjecture 2.3 is phrased in terms of multiples of 6 and the threshold 4206 because Dubner identified the exceptions by exhaustive search and could only describe them by listing them. The sumset of A002822 provides an intrinsic characterisation: an integer belongs to D if and only if it is not reachable from A002822 by addition, with no prior knowledge of D required.

Definition 3.1. $D = \mathbb{N}_{\geq 1} \setminus \{i + j : i, j \in A002822\}$, the complement of the sumset of A002822 in $\mathbb{N}_{\geq 1}$.

By definition, D and $\{i + j : i, j \in A002822\}$ partition $\mathbb{N}_{\geq 1}$. The known value of D up to 2×10^{10} is A243956 [9].

Conjecture 3.2 (Conjecture 2.3, sumset form). The complement of $\{i + j : i, j \in A002822\}$ in $\mathbb{N}_{\geq 1}$ is finite; equivalently, D is finite.

Let T denote the set of t-primes (OEIS A007534 [10]), i.e. primes belonging to a twin prime pair. Every sum of two t-primes is even. Analogously, Conjecture 2.4 has the following sumset reformulation.

Conjecture 3.3 (Conjecture 2.4, sumset form). The complement of $\{p + q : p, q \in T\}$ in the positive even integers is finite.

Remark 3.4. Dubner's computation confirms that every element of D greater than 1 lies in A067611: none of the eleven exceptions beyond 1 is a twin-prime cofactor. The element 1 lies in A002822 (since 5 and 7 are prime) but belongs to D because no two elements of A002822 sum to 1. Once A002822 is given, D is determined by its additive structure alone, without further reference to primality. Note also that Definition 3.1 is robust to the discovery of new exceptions: a new element of D beyond 701 would falsify Conjecture 2.3 (which fixes D as the known finite set) but would merely enlarge D in Definition 3.1, leaving the partition intact.

4. THE BRIDGING CONJECTURE

Conjecture 4.1 (Bridging Conjecture). For every $K \in \mathbb{N}$, there exist $a, b, k \in A002822$ with $a < K$, $b < K$, and $k = a + b > K$.

That is, one can always find two elements of $A002822$ below K whose sum is also in $A002822$ and exceeds K .

Theorem 4.2. *Conjecture 2.3 implies Conjecture 4.1, assuming the Twin Prime Conjecture.*

Proof. Given K , let k be the smallest element of $A002822$ greater than K (which exists by the Twin Prime Conjecture). For large enough K , $6k > 4206$, so Conjecture 2.3 gives $k = a + b$ with $a, b \in A002822$. By minimality of k , both a and b are at most K ; since $a + b = k > K$ they cannot both equal K , so $a, b < K$. $\square$

Theorem 4.3. *Conjecture 2.1 and the Twin Prime Conjecture together imply Conjecture 4.1.*

Proof. Given K , let k be the smallest element of $A002822$ greater than K . By Conjecture 2.1, $k = a + b$ for some $a, b \in A002822$ with $a, b < k$. By minimality of k , we have $a, b \leq K$, and since $a + b > K$ they cannot both equal K , so $a, b < K$. $\square$

Theorem 4.4. *Conjecture 4.1 implies the Twin Prime Conjecture.*

Proof. For any K , the Bridging Conjecture gives $k \in A002822$ with $k > K$. Since K is arbitrary, $A002822$ is infinite. $\square$

Corollary 4.5. *Assuming Conjecture 2.1,*

$$\text{Conjecture 4.1} \iff \text{Twin Prime Conjecture.}$$

Remark 4.6. The role of Conjecture 2.1 in Corollary 4.5 is asymmetric. The forward direction (Bridging Conjecture $\Rightarrow$ Twin Prime Conjecture) holds unconditionally: for any K , the Bridging Conjecture produces $k \in A002822$ with $k > K$, so $A002822$ is unbounded (Theorem 4.4). The reverse direction requires Conjecture 2.1: given that $A002822$ is infinite, let k be the smallest element above K ; Conjecture 2.1 decomposes $k = a + b$ with $a, b \in A002822$, and minimality of k forces $a, b < K$. Without Conjecture 2.1, the Twin Prime Conjecture alone does not guarantee that the next element above K is reachable as a sum of two smaller elements of $A002822$, and the reverse direction fails.

Corollary 4.7. *Conjecture 4.1 implies that for every $K \in \mathbb{N}$ there exists $k \in A002822$ with $K < k < 2K$. Equivalently, consecutive elements $m_n < m_{n+1}$ of $A002822$ satisfy $m_{n+1} < 2m_n$. Consequently, the counting function $\pi_2(x) = |\{m \in A002822 : m \leq x\}|$ satisfies $\pi_2(x) \gg \log x$.*

Proof. The Bridging Conjecture gives $a, b \in A002822$ with $a, b < K$ and $k = a + b \in A002822$ with $k > K$. Since $a, b < K$, we have $k = a + b < 2K$. The gap bound $m_{n+1} < 2m_n$ follows immediately. Since every interval $(m, 2m)$ contains an element of $A002822$, applying this inductively from $m_1 = 1$ gives at least $\lfloor \log_2 x \rfloor$ elements up to x , so $\pi_2(x) \gg \log x$. $\square$

Remark 4.8. Conjecture 2.1 is not required for Corollary 4.7: the bound $k < 2K$ is an immediate consequence of the Bridging Conjecture’s statement alone. What Conjecture 2.1 adds is qualitative: it asserts that *every* element of $A002822$ beyond 1 is reachable as a sum of two smaller elements of $A002822$, not only the specific k produced by the Bridging Conjecture. Note that sums of pairs from $A002822$ need not land in $A002822$ — many land in $A067611$ instead — so this is a non-trivial demand on the self-sumset structure of $A002822$. The upshot is that no element of $A002822$ beyond 1 is an additive dead-end: each one is witnessed by a pair of smaller elements.

5. DISCUSSION

Reformulation of Dubner’s conjectures. Dubner’s original statements of Conjectures 2.3 and 2.4 are phrased in terms of explicit numerical thresholds (4206 and 4208) derived from exhaustive computation. We reformulate them in Conjectures 3.2 and 3.3 as assertions that the complements of the relevant sumsets are finite. This rephrasing has two advantages. First, it removes the dependence on specific threshold values, which are artefacts of how far the computation was carried rather than features of the underlying mathematics. Second, it places Conjectures 2.3 and 2.4 in the same conceptual frame as Definition 3.1: all three are statements about the complement of a sumset in $\mathbb{N}_{\geq 1}$, making the structural parallel between them explicit.

The Bridging Conjecture as a splitting strategy. The Bridging Conjecture offers a way to split the problem. Rather than proving Conjecture 2.3 directly — which requires all sufficiently large multiples of 6 to decompose as sums of two middle numbers — one may instead attempt to establish the two weaker statements independently: Conjecture 2.1 (middle numbers themselves decompose) and the Bridging Conjecture ($A002822$ is unbounded, witnessed by sums of pairs of its own elements). Each of these makes a more focused demand on the additive structure of $A002822$, and either might be more tractable individually than Conjecture 2.3 in full generality.

It is worth noting that Conjecture 2.1 alone does not imply the Twin Prime Conjecture: A002822 could be finite with every element (beyond the smallest) expressible as a sum of two smaller elements, and Conjecture 2.1 would still hold vacuously beyond that point. The Bridging Conjecture supplies exactly what is missing: it asserts that for every K , some pair of elements of A002822 below K sums to an element of A002822 above K , forcing A002822 to be unbounded. In this sense the Bridging Conjecture can be read as a bootstrap principle: if A002822 is known to be non-empty up to K , it certifies the existence of a further element beyond K .

Relationship to D and Conjecture 2.3. The Bridging Conjecture is also strictly weaker than Conjecture 2.3 in a second respect. Conjecture 2.3 makes a precise global claim: D is exactly the finite set A243956. The Bridging Conjecture makes no claim about the size or composition of D — it requires only that the sumset $\{i + j : i, j \in \text{A002822}\}$ is unbounded, which is independent of whether D is finite. It does, however, place one indirect constraint: since the Bridging Conjecture requires infinitely many elements of A002822 to appear as sums of pairs of smaller elements, D cannot eventually absorb all positive integers.

Conditional lower bound on twin prime density. Corollary 4.7 gives a conditional lower bound on the twin prime counting function. The gap bound $m_{n+1} < 2m_n$ implies $\pi_2(x) \gg \log x$ assuming the Bridging Conjecture. For comparison, the best unconditional upper bound is Brun’s theorem [4], which gives $\pi_2(x) \ll x(\log \log x)^2/(\log x)^2$ via sieve methods; this does not establish infinitude. The Hardy–Littlewood conjecture [5] predicts the asymptotic $\pi_2(x) \sim 2C_2 x/(\log x)^2$ where $C_2 \approx 0.6601$ is the twin prime constant, but this remains unproven. Recent unconditional work of Zhang [6] and Maynard [7] establishes infinitely many prime pairs with bounded gaps, but does not yield an explicit lower bound on $\pi_2(x)$. The bound $\pi_2(x) \gg \log x$ from Corollary 4.7 is very weak — far below the Hardy–Littlewood prediction — but it is a consequence of a purely additive conjecture about A002822, with no direct appeal to sieve methods or the distribution of primes in progressions.

Revision history.

- **v1** (17 May 2026): Initial version.
- **v2** (18 May 2026): Revised appendix. The decomposition count figure is replaced by a decomposition gap ratio figure that captures evidence for both Conjecture 1 and Conjecture 3 in a

unified view. The consecutive gap ratio figure is retained. A discussion of the limitations of sieve methods and the additive perspective on the Bridging Conjecture is added.

APPENDIX A. HEURISTIC EVIDENCE

The results of this paper are conditional on unproven conjectures. We present here heuristic arguments and computational evidence suggesting that both Conjecture 2.1 and the Bridging Conjecture are true. These arguments are informal and do not constitute proofs.

Evidence for the Bridging Conjecture. The Bridging Conjecture requires that for every K , some pair of elements of $A002822$ below K sums to an element of $A002822$ above K . By Corollary 4.7, this is equivalent to the gap bound $m_{n+1} < 2m_n$, or equivalently $\text{gap}(n)/m_{n+1} < 1/2$, where $\text{gap}(n) = m_{n+1} - m_n$.

Figure 1 plots $\text{gap}(n)/m_{n+1}$ for elements of $A002822$ up to $n = 10,000$, on a logarithmic scale. The dashed line marks the bound $1/2$. All observed ratios lie well below this bound, and the rolling mean decays steadily toward zero as m grows.

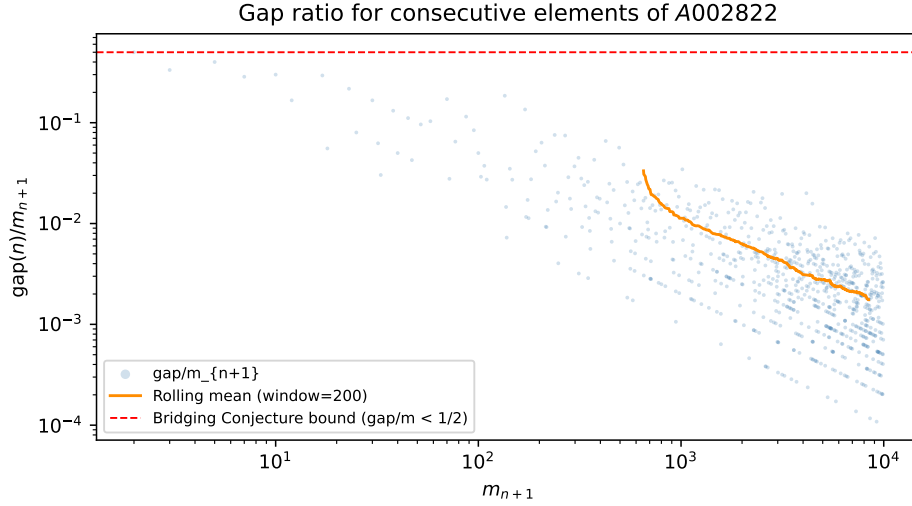


FIGURE 1. Gap ratio $\text{gap}(n)/m_{n+1}$ for consecutive elements of $A002822$ (log scale). The dashed line at $1/2$ marks the bound equivalent to the Bridging Conjecture. The rolling mean decays toward zero, consistent with the Hardy–Littlewood prediction.

Under the Hardy–Littlewood conjecture [5], the density of twin prime cofactors near m is approximately $2C_2/(\log 6m)^2$, so the expected gap between consecutive elements of $A002822$ near m is $\sim (\log 6m)^2/2C_2$. Since $(\log 6m)^2 = o(m)$, the ratio $\text{gap}(n)/m_{n+1} \rightarrow 0$. This is far stronger than the bound of $1/2$ required by the Bridging Conjecture, and explains the rapid decay seen in Figure 1.

Evidence for Conjecture 2.1 and Conjecture 2.3. Figure 2 provides unified evidence for both conjectures. For each $m \in A002822 \cup A243956$, we plot g/m for every decomposition $a + b = m$ with $a \leq b$, where $g = b - a$, on a log-log scale. Elements of D (A243956) have no decomposition within $A002822$; they are assigned the trivial decomposition $(0, m)$, giving $g/m = 1$ always (red crosses, pinned at the dashed line). For $A002822$ elements, all decompositions are shown in blue; the largest-gap decomposition for each m is highlighted in orange.

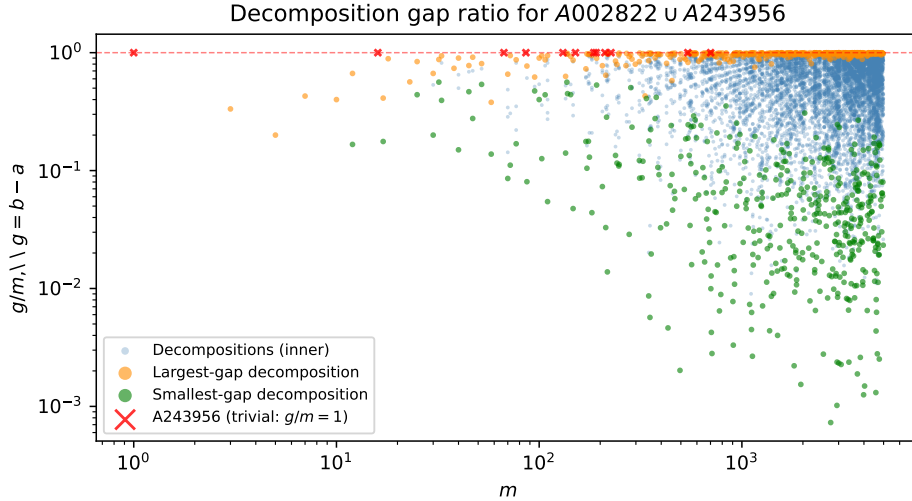


FIGURE 2. Decomposition gap ratio $g/m = (b - a)/m$ for all decompositions $a + b = m$ with $a, b \in A002822$, plotted on a log-log scale. Orange: largest-gap decomposition per m (upper envelope). Green: smallest-gap decomposition per m (lower envelope). Blue: remaining decompositions. Red crosses: elements of D (A243956), using the trivial decomposition $(0, m)$ and pinned at $g/m = 1$. The dashed line marks $g/m = 1$.

Three features of the plot bear directly on the conjectures:

- (1) **No A002822 element above 1 is missing a decomposition** (no blue point is absent for $m > 1$): every element has at least one decomposition, supporting Conjecture 2.1.
- (2) **The D elements are finite and confined to small m :** beyond $m = 701$ no red cross appears, consistent with Dubner's Conjecture 2.3.
- (3) **The orange envelope trends upward toward $g/m = 1$:** the largest-gap decomposition becomes a larger fraction of m as m grows, reflecting that increasingly lopsided decompositions (a, b) with $a \ll b$ become available. The smallest values of g/m among the orange points occur at the smallest elements of A002822, where the only available summands are the two smallest twin prime pairs $(5, 7)$ and $(11, 13)$, corresponding to $m = 1$ and $m = 2$.

The blue cloud fans out on the log-log scale as m increases, reflecting the growing number of decompositions. By analogy with the Goldbach problem, the number of pairs (a, b) with $a + b = m$ and $a, b \in A002822$ is expected to grow as $\sim c \cdot m / (\log m)^2$ by the Hardy–Littlewood circle method, with a constant c reflecting the additional primality constraints on $6a \pm 1$ and $6b \pm 1$. Since this count grows without bound, it is in particular never permanently zero, supporting Conjecture 2.1.

Figure 3 provides a second piece of evidence, directly targeting the $2K$ gap bound of Corollary 4.7. For each $m \in A002822$ with at least two decompositions, it plots $g_{\min}/g_{\max}$, the ratio of the smallest to the largest decomposition gap $g = b - a$. A ratio approaching 1 would mean every decomposition of m is nearly balanced ($a \approx b \approx m/2$), so no pair could bridge far past $K = m/2$ — a direct threat to the gap bound. Counter-examples to the Bridging Conjecture would be expected to appear near the $g_{\min}/g_{\max} = 1$ line. Instead, the ratio trends decisively downward as m grows: the spread between the most and least balanced decompositions increases, and no element comes close to the danger line.

The additive perspective and the limitations of sieve methods.

The density of A002822 is zero: the sieve filters at primes $p \equiv \pm 1 \pmod{6}$ leave a surviving density of $\prod_p (1 - 2/p) \rightarrow 0$. Viewed through this lens, the Bridging Conjecture looks hard — it asks for elements in a set of density zero.

The sieve perspective, however, is ill-suited to the question being asked. Sieve methods assign each prime p a discount factor reflecting its long-run asymptotic effect on density. But the first time p becomes relevant as a factor of $6m \pm 1$ — near $m \sim p/6$ — is a localised event,

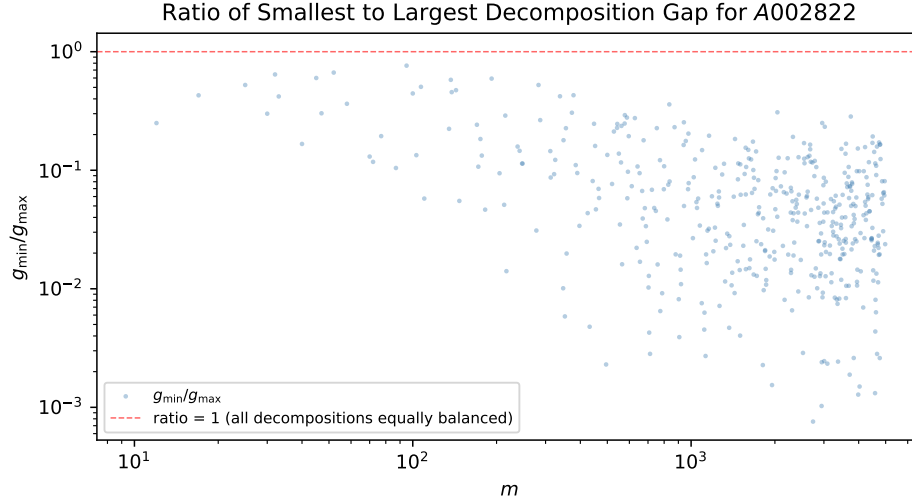


FIGURE 3. Ratio $g_{\min}/g_{\max}$ of smallest to largest decomposition gap for elements of $A002822$ with at least two decompositions, on a log-log scale. The dashed line at 1 marks the dangerous case where all decompositions are equally balanced. The ratio trends downward with m , confirming that the spread of decompositions grows rather than collapses.

not a density. The asymptotic factor $2/p$ amortises this event over all m , which is appropriate for counting survivors globally but obscures the local arithmetic near any particular K .

The Bridging Conjecture is a local question: for a given K , does some pair (a, b) with $a, b < K$ and $a + b \in A002822$ satisfy $a + b > K$? The number of candidate pairs is of order $\pi_2(K)^2/2$, growing like $K^2/(\log K)^4$ under the Hardy–Littlewood conjecture. The probability that a given sum $s = a + b \in (K, 2K)$ lies in $A002822$ is approximately $2C_2/(\log K)^2$, so the expected witness count grows like $K^2/(\log K)^6$ — without bound. Moreover, the joint sieve constraints on the triple $(a, b, a + b)$ are positively correlated: a residue class that is “safe” for a and b modulo a prime p tends also to be safe for $a + b$. This means the true singular series constant exceeds the naive $(2C_2)^3$ estimate by a large factor (empirically around 49), so the arithmetic structure actively favours witnesses.

From the additive perspective, the conjecture is not merely plausible but overwhelmingly supported. The remaining gap between heuristic and proof is correlation control: showing that the many candidate

witness pairs do not conspire to fail simultaneously. This is the same obstruction that keeps Goldbach’s conjecture open despite its heuristic being overwhelming, and it is not addressed by sieve methods. Whether additive combinatorics tools — the circle method, Fourier analysis on $\mathbb{Z}/N\mathbb{Z}$, density increment arguments — can make progress here that sieve methods cannot is an open question, but the framing itself suggests that asymptotic sieve techniques may be directed at the wrong structure.

ACKNOWLEDGEMENTS

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